SUPPLEMENTAL MATERIAL

Data S1.

Detailed Search Strategies

Search Strategy 1

PubMed:

("NOACs"[tw] OR "NOAC"[tw] OR "DOACs"[tw] OR "DOAC"[tw] OR "new oral anticoagulant" [tw] OR "new oral anti coagulant"[tw] OR "novel oral anticoagulant"[tw] OR "new oral anti coagulant"[tw] OR "new oral anticoagulants"[tw] OR "new oral anti coagulants"[tw] OR "novel oral anticoagulants"[tw] OR "new oral anti coagulants"[tw] OR "non-vitamin K antagonist oral anticoagulants"[tw] OR "non-vitamin K antagonist oral anticoagulant"[tw] OR "novel oral anticoagulants"[tw] OR "new oral anticoagulants"[tw] OR "direct oral anticoagulants"[tw] OR "direct oral anticoagulant"[tw] OR "direct oral anti coagulant"[tw] OR "direct oral anti coagulants"[tw] OR "factor Xa inhibitors"[tw] OR "factor IIa inhibitors"[tw] OR "direct thrombin inhibitor"[tw] OR "direct thrombin inhibitors"[tw] OR "factor xa inhibitor"[tw] OR "factor IIa inhibitor"[tw] OR "dabigatran"[tw] OR "Pradaxa"[tw] OR "rivaroxaban"[tw] OR "Xarelto"[tw] OR "apixaban"[tw] OR "Eliquis"[tw] OR "eliques"[tw] OR "edoxaban"[tw] OR "Savaysa"[tw] OR "roteas"[tw] OR “Lixiana” [tw] OR “betrixaban”[tw] OR “Bevyxxa”[tw] OR "edoxaban" [Supplementary Concept] OR "apixaban" [Supplementary Concept] OR "Dabigatran"[Mesh] OR "Rivaroxaban"[Mesh] OR "betrixaban" [Supplementary Concept] OR (("new oral"[tw] OR "novel"[tw] OR "non vitamin k"[tw]) AND (anticoagulant*[tw] OR anti coagulant*[tw])))

AND

("warfarin" OR "vitamin k" OR "Coumadin" OR "acenocoumarol" OR "acenocoumarol"[mh] OR "phenprocoumon"[mh] OR "phenprocoumon" OR "Vitamin K/antagonists and inhibitors"[Mesh] OR "jantoven")

AND

("fracture" OR "fractures" OR "fractures, bone"[mh])

Limit: English

EMBASE:

('NOACs' OR 'NOAC' OR 'DOACs' OR 'DOAC' OR 'new oral anticoagulant' OR 'new oral anti coagulant' OR 'novel oral anticoagulant' OR 'new oral anti coagulant' OR 'new oral anticoagulants' OR 'new oral anti coagulants' OR 'novel oral anticoagulants' OR 'new oral anti coagulants' OR 'non-vitamin K antagonist oral anticoagulants' OR 'non-vitamin K antagonist oral anticoagulant' OR 'novel oral anticoagulants' OR 'new oral anticoagulants' OR 'direct oral anticoagulants' OR 'direct oral anticoagulant' OR 'direct oral anti coagulant' OR 'direct oral anti coagulants' OR 'factor Xa inhibitors' OR 'factor IIa inhibitors' OR 'direct thrombin inhibitor' OR 'direct thrombin inhibitors' OR 'factor xa inhibitor' OR 'factor IIa inhibitor' OR 'dabigatran' OR 'Pradaxa' OR 'rivaroxaban' OR 'Xarelto' OR 'apixaban' OR 'Eliquis' OR 'eliques' OR 'edoxaban' OR 'Savaysa' OR 'roteas' OR 'Lixiana' OR 'betrixaban' OR 'Bevyxxa' OR 'edoxaban'/exp OR 'apixaban'/exp OR 'Dabigatran'/exp OR 'Rivaroxaban'/exp OR 'betrixaban'/exp OR (('new oral' OR 'novel' OR 'non vitamin k') AND (anticoagulant* OR anti coagulant*)))

AND

('warfarin' OR 'vitamin k' OR 'Coumadin' OR 'acenocoumarol' OR 'acenocoumarol'/exp OR 'phenprocoumon'/exp OR 'phenprocoumon' OR 'antivitamin K'/exp OR 'jantoven')

AND

('fracture' OR 'fractures' OR 'fracture'/exp)

Limit: English

Cochrane Library:

("NOACs":ti,ab,kw OR "NOAC":ti,ab,kw OR "DOACs":ti,ab,kw OR "DOAC":ti,ab,kw OR "new oral anticoagulant":ti,ab,kw OR "new oral anti coagulant":ti,ab,kw OR "novel oral anticoagulant":ti,ab,kw OR "new oral anti coagulant":ti,ab,kw OR "new oral anticoagulants":ti,ab,kw OR "new oral anti coagulants":ti,ab,kw OR "novel oral anticoagulants":ti,ab,kw OR "new oral anti

coagulants":ti,ab,kw OR "non vitamin K antagonist oral anticoagulants":ti,ab,kw OR "non vitamin K antagonist oral anticoagulant":ti,ab,kw OR "novel oral anticoagulants":ti,ab,kw OR "new oral anticoagulants":ti,ab,kw OR "direct oral anticoagulants":ti,ab,kw OR "direct oral anticoagulant":ti,ab,kw OR "direct oral anti coagulant":ti,ab,kw OR "direct oral anti coagulants":ti,ab,kw OR "factor Xa inhibitors":ti,ab,kw OR "factor IIa inhibitors":ti,ab,kw OR "direct thrombin inhibitor":ti,ab,kw OR "direct thrombin inhibitors":ti,ab,kw OR "factor xa inhibitor":ti,ab,kw OR "factor IIa inhibitor":ti,ab,kw OR "dabigatran":ti,ab,kw OR "Pradaxa":ti,ab,kw OR "rivaroxaban":ti,ab,kw OR "Xarelto":ti,ab,kw OR "apixaban":ti,ab,kw OR "Eliquis":ti,ab,kw OR "eliques":ti,ab,kw OR "edoxaban":ti,ab,kw OR "Savaysa":ti,ab,kw OR "roteas":ti,ab,kw OR "Lixiana":ti,ab,kw OR "betrixaban":ti,ab,kw OR "Bevyxxa":ti,ab,kw OR [mh "dabigatran"] OR [mh "rivaroxaban"] OR [mh "apixaban"] OR [mh "edoxaban"] OR [mh "betrixaban"] OR (("new oral":ti,ab,kw OR "novel":ti,ab,kw OR "non vitamin k":ti,ab,kw) AND (anticoagulant*:ti,ab,kw OR anti coagulant*:ti,ab,kw)))

AND

("vitamin K antagonists":ti,ab,kw OR "warfarin":ti,ab,kw OR "Coumadin":ti,ab,kw OR "Acenocoumarol":ti,ab,kw OR "Phenprocoumon":ti,ab,kw OR "Phenprocoumon":ti,ab,kw)

AND

("Fracture":ti,ab,kw)

Scopus:

TITLE-ABS-KEY(({NOACs} OR {NOAC} OR {DOACs} OR {DOAC} OR {new oral anticoagulant} OR {new oral anti coagulant} OR {novel oral anticoagulant} OR {new oral anti coagulant} OR {new oral anticoagulants} OR {new oral anti coagulants} OR {novel oral anticoagulants} OR {new oral anti coagulants} OR {non-vitamin K antagonist oral anticoagulants} OR {non-vitamin K antagonist oral anticoagulant} OR {novel oral anticoagulants} OR {new oral anticoagulants} OR {direct oral anticoagulants} OR {direct oral

anticoagulant} OR {direct oral anti coagulant} OR {direct oral anti coagulants} OR {factor Xa inhibitors} OR {factor IIa inhibitors} OR {direct thrombin inhibitor} OR {direct thrombin inhibitors} OR {factor xa inhibitor} OR {factor IIa inhibitor} OR {dabigatran} OR {Pradaxa} OR {rivaroxaban} OR {Xarelto} OR {apixaban} OR {Eliquis} OR {eliques} OR {edoxaban} OR {Savaysa} OR {roteas} OR {Lixiana} OR {betrixaban} OR {Bevyxxa} OR (({new oral} OR {novel} OR {non vitamin k}) AND (anticoagulant* OR anti coagulant*)))

AND

({warfarin} OR {vitamin k} OR {Coumadin} OR {acenocoumarol} OR {phenprocoumon} OR {jantoven})

AND

({fracture} OR {fractures}))

Limit: English

Web of Science:

TS=(("NOACs" OR "NOAC" OR "DOACs" OR "DOAC" OR "new oral anticoagulant" OR "new oral anti coagulant" OR "novel oral anticoagulant" OR "new oral anti coagulant" OR "new oral anticoagulants" OR "new oral anti coagulants" OR "novel oral anticoagulants" OR "new oral anti coagulants" OR "non-vitamin K antagonist oral anticoagulants" OR "non-vitamin K antagonist oral anticoagulant" OR "novel oral anticoagulants" OR "new oral anticoagulants" OR "direct oral anticoagulants" OR "direct oral anticoagulant" OR "direct oral anti coagulant" OR "direct oral anti coagulants" OR "factor Xa inhibitors" OR "factor IIa inhibitors" OR "direct thrombin inhibitor" OR "direct thrombin inhibitors" OR "factor xa inhibitor" OR "factor IIa inhibitor" OR "dabigatran" OR "Pradaxa" OR "rivaroxaban" OR "Xarelto" OR "apixaban" OR "Eliquis" OR "eliques" OR "edoxaban" OR "Savaysa" OR "roteas" OR "Lixiana" OR "betrixaban" OR "Bevyxxa" OR (("new oral" OR "novel" OR "non vitamin k") AND (anticoagulant* OR anti coagulant*)))

AND

("warfarin" OR "vitamin k" OR "Coumadin" OR "acenocoumarol" OR "phenprocoumon" OR "jantoven")

AND

("fracture" OR "fractures"))

Limit: English

Search Strategy 2

PubMed:

("NOACs"[tw] OR "NOAC"[tw] OR "DOACs"[tw] OR "DOAC"[tw] OR "new oral anticoagulant" [tw] OR "new oral anti coagulant"[tw] OR "novel oral anticoagulant"[tw] OR "new oral anti coagulant"[tw] OR "new oral anticoagulants"[tw] OR "new oral anti coagulants"[tw] OR "novel oral anticoagulants"[tw] OR "new oral anti coagulants"[tw] OR "non-vitamin K antagonist oral anticoagulants"[tw] OR "non-vitamin K antagonist oral anticoagulant"[tw] OR "novel oral anticoagulants"[tw] OR "new oral anticoagulants"[tw] OR "direct oral anticoagulants"[tw] OR "direct oral anticoagulant"[tw] OR "direct oral anti coagulant"[tw] OR "direct oral anti coagulants"[tw] OR "factor Xa inhibitors"[tw] OR "factor IIa inhibitors"[tw] OR "direct thrombin inhibitor"[tw] OR "direct thrombin inhibitors"[tw] OR "factor xa inhibitor"[tw] OR "factor IIa inhibitor"[tw] OR "dabigatran"[tw] OR "Pradaxa"[tw] OR "rivaroxaban"[tw] OR "Xarelto"[tw] OR "apixaban"[tw] OR "Eliquis"[tw] OR "eliques"[tw] OR "edoxaban"[tw] OR "Savaysa"[tw] OR "roteas"[tw] OR “Lixiana” [tw] OR “betrixaban”[tw] OR “Bevyxxa”[tw] OR "edoxaban" [Supplementary Concept] OR "apixaban" [Supplementary Concept] OR "Dabigatran"[Mesh] OR "Rivaroxaban"[Mesh] OR "betrixaban" [Supplementary Concept] OR (("new oral"[tw] OR "novel"[tw] OR "non vitamin k"[tw]) AND (anticoagulant*[tw] OR anti coagulant*[tw])))
AND
(randomized controlled trial [pt] OR controlled clinical trial [pt] OR randomized [tiab] OR randomised [tiab] OR placebo [tiab] OR randomly [tiab] OR trial [tiab]) NOT (“animals”[mesh] NOT (“animals”[mesh] AND “humans”[mesh]))
Limits: English

EMBASE:

('NOACs':ti,ab,kw OR 'NOAC':ti,ab,kw OR 'DOACs':ti,ab,kw OR 'DOAC':ti,ab,kw OR 'new oral anticoagulant':ti,ab,kw OR 'new oral anti coagulant':ti,ab,kw OR 'novel oral anticoagulant':ti,ab,kw OR 'new oral anti coagulant':ti,ab,kw OR 'new oral

anticoagulants':ti,ab,kw OR 'new oral anti coagulants':ti,ab,kw OR 'novel oral anticoagulants':ti,ab,kw OR 'new oral anti coagulants':ti,ab,kw OR 'non-vitamin K antagonist oral anticoagulants':ti,ab,kw OR 'non-vitamin K antagonist oral anticoagulant':ti,ab,kw OR 'novel oral anticoagulants':ti,ab,kw OR 'new oral anticoagulants':ti,ab,kw OR 'direct oral anticoagulants':ti,ab,kw OR 'direct oral anticoagulant':ti,ab,kw OR 'direct oral anti coagulant':ti,ab,kw OR 'direct oral anti coagulants':ti,ab,kw OR 'factor Xa inhibitors':ti,ab,kw OR 'factor IIa inhibitors':ti,ab,kw OR 'direct thrombin inhibitor':ti,ab,kw OR 'direct thrombin inhibitors':ti,ab,kw OR 'factor xa inhibitor':ti,ab,kw OR 'factor IIa inhibitor':ti,ab,kw OR 'dabigatran':ti,ab,kw,de OR 'Pradaxa':ti,ab,kw,de OR 'rivaroxaban':ti,ab,kw,de OR 'Xarelto':ti,ab,kw,de OR 'apixaban':ti,ab,kw,de OR 'Eliquis':ti,ab,kw,de OR 'eliques':ti,ab,kw,de OR 'edoxaban':ti,ab,kw,de OR 'Savaysa':ti,ab,kw,de OR 'roteas':ti,ab,kw,de OR 'Lixiana':ti,ab,kw,de OR 'betrixaban':ti,ab,kw,de OR 'Bevyxxa':ti,ab,kw,de OR (('new oral':ti,ab,kw OR 'novel':ti,ab,kw OR 'non vitamin k':ti,ab,kw) AND (anticoagulant*:ti,ab,kw OR anti coagulant*:ti,ab,kw)))

AND

('randomized controlled trial'/de OR 'controlled clinical trial'/de OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab) NOT ('animal'/exp NOT ('animal'/exp AND 'human'/exp))

Limits: English

Cochrane Library:

("NOACs":ti,ab,kw OR "NOAC":ti,ab,kw OR "DOACs":ti,ab,kw OR "DOAC":ti,ab,kw OR "new oral anticoagulant":ti,ab,kw OR "new oral anti coagulant":ti,ab,kw OR "novel oral anticoagulant":ti,ab,kw OR "new oral anti coagulant":ti,ab,kw OR "new oral anticoagulants":ti,ab,kw OR "new oral anti coagulants":ti,ab,kw OR "novel oral anticoagulants":ti,ab,kw OR "new oral anti coagulants":ti,ab,kw OR "non vitamin K antagonist oral anticoagulants":ti,ab,kw OR "non vitamin K antagonist oral anticoagulant":ti,ab,kw OR "novel oral anticoagulants":ti,ab,kw OR "new oral anticoagulants":ti,ab,kw OR "direct oral

anticoagulants":ti,ab,kw OR "direct oral anticoagulant":ti,ab,kw OR "direct oral anti coagulant":ti,ab,kw OR "direct oral anti coagulants":ti,ab,kw OR "factor Xa inhibitors":ti,ab,kw OR "factor IIa inhibitors":ti,ab,kw OR "direct thrombin inhibitor":ti,ab,kw OR "direct thrombin inhibitors":ti,ab,kw OR "factor xa inhibitor":ti,ab,kw OR "factor IIa inhibitor":ti,ab,kw OR "dabigatran":ti,ab,kw OR "Pradaxa":ti,ab,kw OR "rivaroxaban":ti,ab,kw OR "Xarelto":ti,ab,kw OR "apixaban":ti,ab,kw OR "Eliquis":ti,ab,kw OR "eliques":ti,ab,kw OR "edoxaban":ti,ab,kw OR "Savaysa":ti,ab,kw OR "roteas":ti,ab,kw OR "Lixiana":ti,ab,kw OR "betrixaban":ti,ab,kw OR "Bevyxxa":ti,ab,kw OR [mh "dabigatran"] OR [mh "rivaroxaban"] OR [mh "apixaban"] OR [mh "edoxaban"] OR [mh "betrixaban"] OR (("new oral":ti,ab,kw OR "novel":ti,ab,kw OR "non vitamin k":ti,ab,kw) AND (anticoagulant*:ti,ab,kw OR anti coagulant*:ti,ab,kw)))

ClinicalTrials.gov:

"new oral anticoagulant" OR "novel oral anticoagulant" OR "non vitamin K antagonist oral anticoagulant" OR "direct oral anticoagulant" OR “NOAC” OR “DOAC” OR "dabigatran" OR "apixaban" OR "edoxaban" OR "rivaroxaban" OR "betrixaban"

Table S1. Summary of the included cohort studies in the present meta-analysis

Author name	Year	Design	Country	Treatment indication	NOAC type	Mean Follow-up (months)	Mean age (years)	Female (%)	Study sample size (n)	Reported fracture sites
Lucenteforte[12]	2017	Retrospective cohort study	Italy	Not specified	Dabigatran and direct Xa inhibitors (combined rivaroxaban and apixaban)	<12*	Not provided‡	48.9	16,850	Hip or vertebral fracture
Binding[14]	2019	Retrospective cohort study	Denmark	AF	Dabigatran, rivaroxaban, apixaban, and edoxaban (all combined)	24	72.7	42.2	37,350	Any fracture; major osteoporotic fracture; hip fracture
Huang[16]	2020	Retrospective cohort study	Taiwan	AF	Dabigatran, rivaroxaban, and apixaban	29.2†	71.9	41.0	28,776§	Hip; vertebral; and humerus/forearm/wrist fractures
Lau[17]	2020	Retrospective cohort study	Hong Kong	AF	Dabigatran, rivaroxaban, and apixaban	14.1	74.4	48.0	23,515	Hip or vertebral fracture
Lutsey[15]	2020	Retrospective cohort study	USA	AF	Dabigatran, rivaroxaban, and apixaban	16.9	68.9	38.0	167,275§	Hip fracture and all clinical fractures

* no report of the actual mean follow-up time.

† Median.

‡ 67.2% patients ≥75 years old; 32.8% patients <75 years old.

§ Overall study sample size before head-to-head propensity score matching.

Abbreviations: AF, atrial fibrillation; NOAC, non-vitamin K antagonist oral anticoagulant; OAC, oral anticoagulant

Table S2. Summary of the included randomized controlled trials in the present meta-analysis

Trial name	Year	Country	Treatment indication	NOAC type	Follow-up duration (months)	Mean age (years)	Female (%)	Study sample size (n)	Reported fracture sites
PETRO Study[30]	2007	4 (Denmark, Netherlands, Sweden, and USA)	AF	Dabigatran	3	69.7	18.1	502	Radius
RE-COVER I[31]	2009	29 countries	VTE	Dabigatran	6	54.7	41.6	2,539	Femur; hip; lower limb; radius; rib; tibia
RE-LY[32]	2009	44 countries	AF	Dabigatran	24*	71	36.4	18,040	Ankle; cervical vertebral; clavicle; compression; facial bones; femoral neck; femur; fibula; foot; forearm; hand; hip; humerus; lower limb; lumbar vertebral; multiple; patella; pelvic; pubic rami; radius; rib; scapula; skull; spinal compression; spinal; sternal; thoracic vertebral; tibia; upper limb; wrist; pathological; unspecified
EINSTEIN-DVT‖[33]	2010	35 countries	VTE	Rivaroxaban	3, 6, or 12†	56.1	43.2	3,429	Ankle; clavicle; femoral neck; femur; humerus; radius; rib; spinal compression; ulna; thoracic vertebral; pathological
NCT00504556‡[34]	2010	12 countries	AF	Edoxaban	4	65	37.9	1,143	Clavicle; upper limb
NCT01136408‡[35]	2010	1 (Japan)	AF	Dabigatran	3	67.8	12.3	166	Femur
ARISTOTLE[36]	2011	39 countries	AF	Apixaban	21.6*	Median 70	35.3	18,140	Hand; periprosthetic; upper limb; foot; spinal compression; facial bones; open; tibia; acetabulum; ankle; clavicle; femoral neck; pelvic; scapula; skull; wrist; cervical vertebral; femur; hip; lower limb; patella; thoracic vertebral; pubis; spinal; sternal; fibula; forearm; unspecified; humerus; rib; traumatic; lumbar vertebral; radius; ulna; pathological
NCT00806624‡[37]	2011	4 (Taiwan, South Korea, Hong Kong, and Singapore)	AF	Edoxaban	3	65.1	34.6	234	Unspecified fracture
ROCKET AF[38]	2011	45 countries	AF	Rivaroxaban	23.6*	Median 73	39.7	14,236	Ankle; avulsion; cervical vertebral; clavicle;

									compression; facial bones; femoral neck; femur; fibula; foot; unspecified; hand; hip; humerus; lower limb; lumbar vertebral; multiple; patella; pelvic; pubis; radius; rib; skull; spinal compression; spinal; thoracic vertebral; tibia; traumatic; upper limb; wrist; pathological
EINSTEIN-PE[‖][39]	2012	38 countries	VTE	Rivaroxaban	3, 6, or 12[†]	57.7	47.1	4,817	Ankle; facial bones; femoral neck; femur; fibula; foot; hip; humerus; rib; spinal compression; sternal; tibia; traumatic; cervical vertebral; lumbar vertebral; thoracic vertebral; upper limb
J-ROCKET AF[40]	2012	1 (Japan)	AF	Rivaroxaban	16.2*	71.1	19.4	1,278	Femur; fibula; patella; radius; rib; spinal compression; tibia; ulna; skull
AMPLIFY[41]	2013	28 countries	VTE	Apixaban	at least 6	56.9	41.3	5,365	Ankle; cervical vertebral; facial bones; femur; hip; humerus; lower limb; lumbar vertebral; pelvic; radius; spinal compression; upper limb; wrist; pathological
ENGAGE AF-TIMI 48[42]	2013	46 countries	AF	Edoxaban	33.6*	Median 72	38.1	21,026	Femur; hip; rib; humerus; spinal compression; femoral neck; lower limb; lumbar vertebral; facial bones; pelvic; fibula; thoracic vertebral; ankle; pubis; cervical vertebral; foot; forearm; patella; radius; ulna; wrist; acetabulum; sacrum; hand; periprosthetic; spinal; tibia; upper limb; clavicle; compression; jaw; multiple; scapula; skull; unspecified; coccyx; ischium; skull; sternal; osteoporotic; pathological
Hokusai-VTE[43]	2013	37 countries	VTE	Edoxaban	12	55.8	42.8	8,240	Femur; rib; spinal compression; femoral neck; tibia; acetabulum; hip; jaw; radius; spinal; ankle; clavicle; compression; fibula; foot; sacrum; hand; humerus; lower limb; pelvic; pubis; thoracic vertebral; upper limb; wrist; facial bones; forearm; lumbar vertebral; multiple; traumatic
RE-COVER II[44]	2013	31 countries	VTE	Dabigatran	6	54.9	39.4	2,568	Femoral neck; hip; humerus; multiple; upper limb
RE-MEDY[45]	2013	33 countries	VTE	Dabigatran	6-36	54.7	39	2,856	Acetabulum; ankle; femoral neck; femur; fibula; foot; hand; hip; humerus; lower limb; radius; tibia; upper limb
ENSURE-AF[46]	2016	19 countries	AF	Edoxaban	2	64.2	34.4	2,149	Ankle

eTRIS[47]	2016	1 (USA)	VTE	Edoxaban	4	54.8	26.2	84	Femoral neck; rib
PIONEER AF-PCI[48]	2016	14 countries	AF	Rivaroxaban	12	70.1	25.5	2,099	Compression; femoral neck; hip; humerus; lower limb; pubis; radius; rib; traumatic; upper limb
RE-CIRCUIT[49]	2017	11 countries	AF	Dabigatran	3-4	59.2	25.2	676	Acetabulum
REDUAL-PCI[50]	2017	41 countries	AF	Dabigatran	14.0[§]	70.8	24	2,678	Ankle; clavicle; femoral neck; femur; hand; hip; lumbar vertebral; multiple; pelvic; periprosthetic; pubis; rib; skull; spinal compression; sternal; upper limb; wrist
ELIMINATE-AF[‖][51]	2018	11 countries	AF	Edoxaban	5	59.5	28.1	602	Foot
EMANATE[52]	2018	12 countries	AF	Apixaban	1-4	64.6	33.2	1,456	Femur; radius
RE-SPECT CVT[53]	2019	9 countries	VTE	Dabigatran	6	45.2	55	120	Upper limb

*****Median.

[†]Depended on the intended duration of treatment.

[‡]Named by clinical trial registry number.

[§]Median.

[‖]The control group in these studies contained both warfarin and acenocoumarol users; the present meta-analysis included them in the warfarin group.

Abbreviations: AF, atrial fibrillation; NOAC, non-vitamin K antagonist oral anticoagulant; OAC, oral anticoagulant; VTE, venous thromboembolism

Table S3. Quality assessment of the included cohort studies using the Newcastle–Ottawa Scale

Author	Year	Selection				Comparability		Outcome			Total Quality Score
		Representativeness of exposed cohort	Selection of non-exposed cohort	Ascertainment of exposure	The outcome of interest was not present at the start of the study	Adjusted for age	Adjusted for sex	Assessment of outcome	Follow-up length	Loss to follow-up rate	
Lucenteforte	2017	1	1	0	1	1	1	1	0	0	**6**
Binding	2019	1	1	1	1	1	1	1	1	1	**9**
Huang	2020	1	1	1	1	1	1	1	1	1	**9**
Lau	2020	1	1	1	1	1	1	1	1	1	**9**
Lutsey	2020	1	1	1	1	1	1	1	1	1	**9**

Selection:

1) Representativeness of exposed cohort: 1, truly or somewhat representative of a community/population-based study; 0, selected group of users or no description of the derivation of the cohort.

2) Selection of non-exposed cohort: 1, drawn from the same community as the exposed cohort; 0, drawn from a different source or no description of the derivation of the non-exposed cohort.

3) Ascertainment of exposure: 1, validation of oral anticoagulant use with a secure record or structured interview; 0, written self-report or no description of validation of oral anticoagulant use.

4) Demonstration that outcome of interest was not present at the start of the study: 1, yes; 0, no.

Comparability:

1) Study adjusted for age: 1, yes; 0, no.

2) Study adjusted for sex: 1, yes; 0, no.

Outcome:

1) Assessment of outcome: 1, independent blind assessment, confirmed by medical records or record linkage; 0, self-reported or no description.

2) Was follow-up long enough for outcomes to occur: 1, duration of follow-up ≥ 1 year; 0, duration of follow-up < 1 year.

3) Loss to follow-up rate: 1, complete follow-up or loss to follow up rate ≤20%; 0, loss to follow up rate >20% or no statement.

Table S4. Quality assessment of the included randomized controlled trials using the Cochrane risk of bias tool

Trial name	Year	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data addressed	Selective reporting	Other biases
PETRO Study	2007	U	U	H	L	L	L	L
RE-COVER I	2009	L	L	L	L	L	L	L
RE-LY	2009	L	L	H	L	L	L	L
EINSTEIN-DVT	2010	L	L	H	L	L	L	L
NCT00504556	2010	L	L	H	L	L	L	L
NCT01136408	2010	L	U	H	U	U	L	U
ARISTOTLE	2011	L	L	L	L	L	L	L
NCT00806624	2011	L	L	H	L	L	L	L
ROCKET AF	2011	L	L	H	L	L	L	L
EINSTEIN-PE	2012	L	L	H	L	L	L	L
J-ROCKET AF	2012	U	U	L	L	U	L	L
AMPLIFY	2013	L	L	L	L	L	L	L
ENGAGE AF-TIMI 48	2013	L	L	L	L	L	L	L
Hokusai-VTE	2013	L	L	L	L	L	L	L
RE-COVER II	2013	L	L	L	L	L	L	L
RE-MEDY	2013	L	L	L	L	U	L	L
ENSURE-AF	2016	L	L	H	L	L	L	L
eTRIS	2016	L	L	H	L	U	L	L
PIONEER AF-PCI	2016	L	L	H	U	L	L	L
RE-CIRCUIT	2017	L	L	H	L	L	L	L
REDUAL-PCI	2017	L	L	H	L	L	L	L
ELIMINATE-AF	2018	L	L	H	L	L	L	L

EMANATE	2018	L	L	H	U	L	L	L
RE-SPECT CVT	2019	L	L	H	L	L	L	L

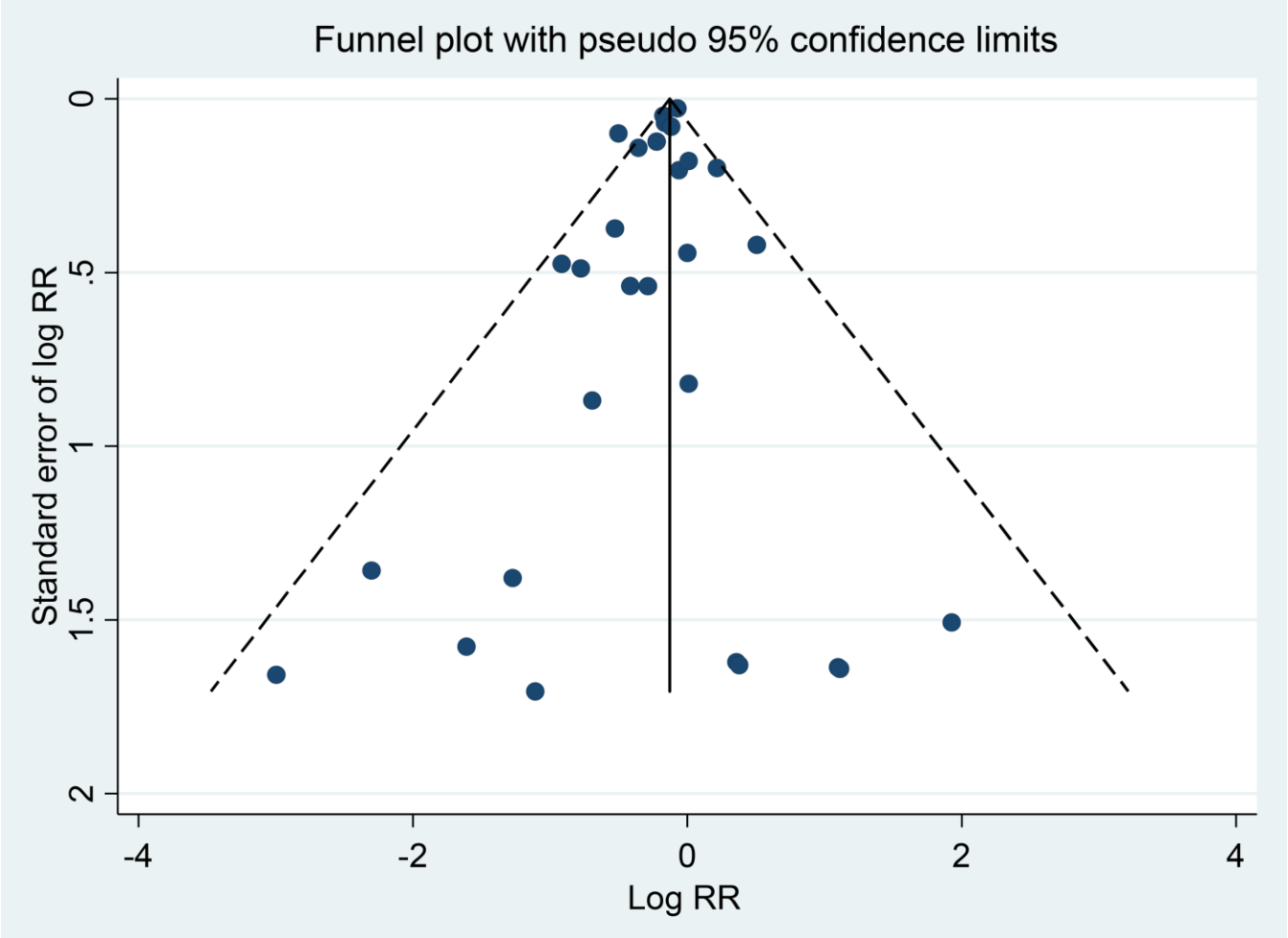
L: low risk; U: unclear risk; H: high risk

Table S5. Covariates adjusted in each cohort study included in the present meta-analysis

Author	Year	Age	Sex	HTN	DM	COPD	CAD/MI	Heart failure	CKD	Stroke	Dementia	Osteoporosis/ BMD	Medication use	Other variables
Lucenteforte	2017	●	●											The pattern of OAC use (incident or non-incident)
Binding	2019	●	●		●	●		●		●			Hormone replacement therapy, antidepressant drugs, glucocorticoid medication, statin	Previous syncope, liver disease, inflammatory polyarthritis, alcohol
Huang	2020	●	●	●	●	●	●	●	●	●	●	●	Corticosteroids, diuretics, NSAID, statins, PPI, antiepileptics, antiparkinsonian, antipsychotics, anxiolytics, hypnotics and sedatives, antidepressants, thyroxine, antithyroid drugs, antiosteoporotic drugs	Thyroid disease, rheumatoid arthritis, malignancy, income level, time from AF diagnosis to OAC prescription, hospital level, hospital region, physician specialty, Charlson comorbidity index, cirrhosis, depression, parkinsonism, epilepsy, cataract, CHA2DS2-VASc score
Lau	2020	●	●	●	●	●		●	●	●		●	ACEI, ARB, β-Blockers, PPI, bisphosphonates, antidepressants, systemic glucocorticoids	Previous fall, previous fracture, rheumatoid arthritis, liver disease, inflammatory polyarthropathies
Lutsey	2020	●	●	●	●	●	●	●	●	●	●	●	Digoxin, clopidogrel, antiplatelets, ACEI, ARB, β-blockers, calcium channel blockers, antiarrhythmic agents, statins, diabetes medications, osteoporosis medications	Previous fracture, peripheral artery disease, liver disease, depression, hematological disorders, GI tract bleeding, other bleeding, frailty, malignancy, prior procedures (cardiac, vascular, GI tract, and neurological), alcohol, CHA2DS2-VASc score

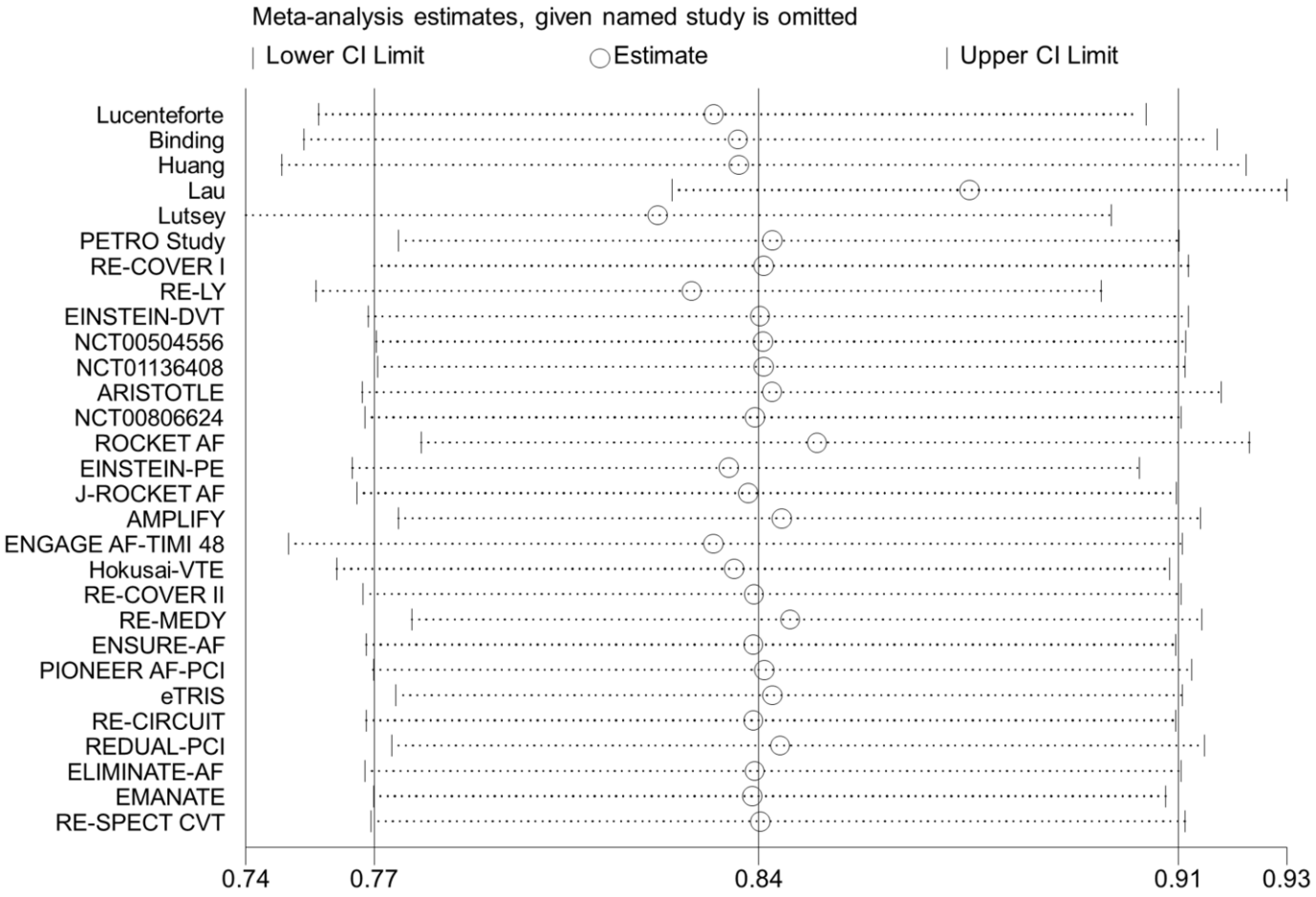
Abbreviations: ACEI, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; AF, atrial fibrillation; BMD, bone mineral density; CAD, coronary artery disease; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; DM, diabetes mellitus; GI, gastrointestinal; HTN, hypertension; MI, myocardial infarction; NSAID, nonsteroidal anti-inflammatory drugs; OAC, oral anticoagulants; PPI, proton-pump inhibitors

Figure S1. Funnel plot for the publication bias in studies comparing any fracture risk between NOACs and warfarin



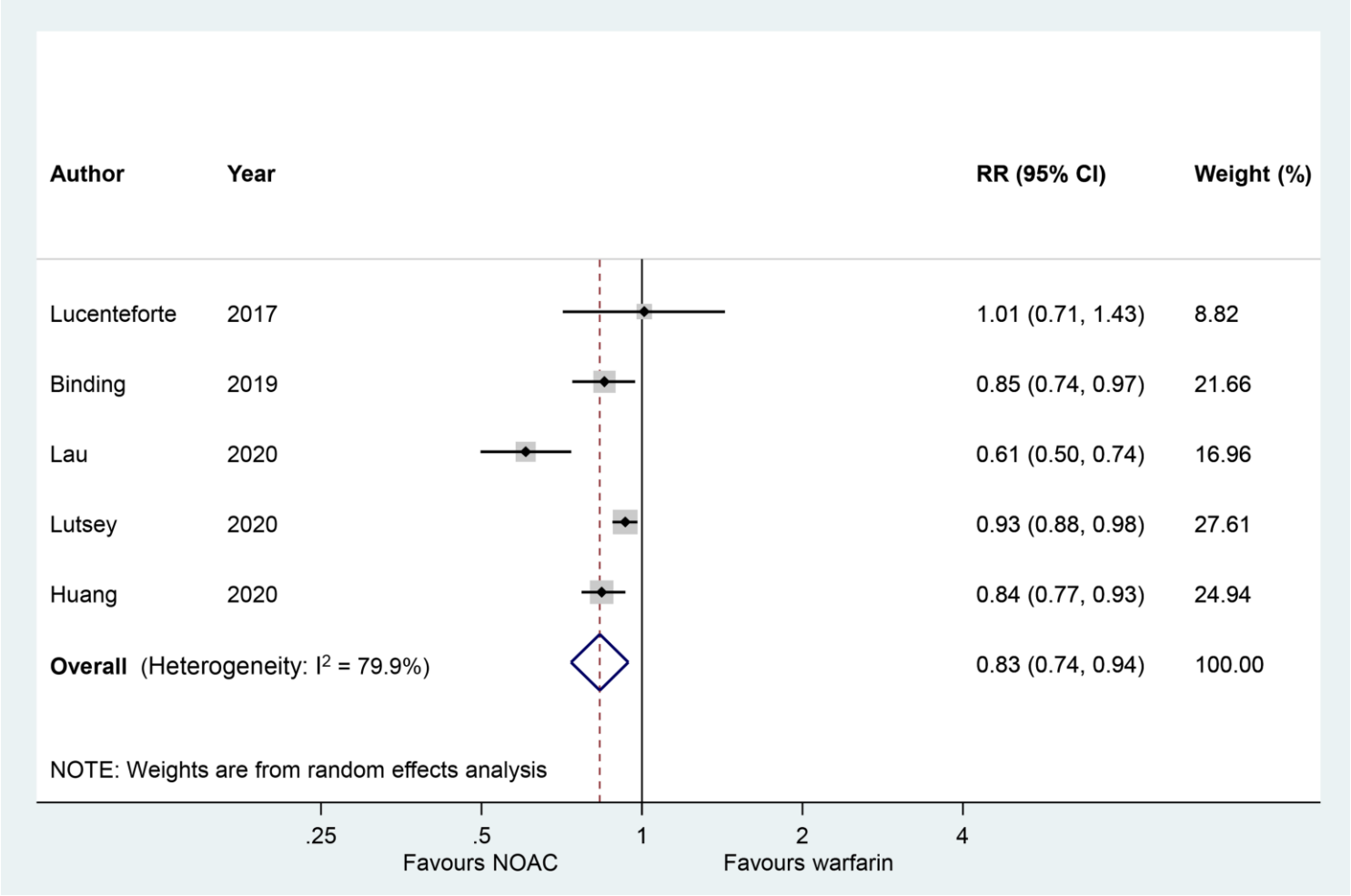
Abbreviations: RR, relative risk

Figure S2. Sensitivity analysis of the meta-analysis of studies comparing any fracture risk between NOACs and warfarin with each study omitted individually one at a time



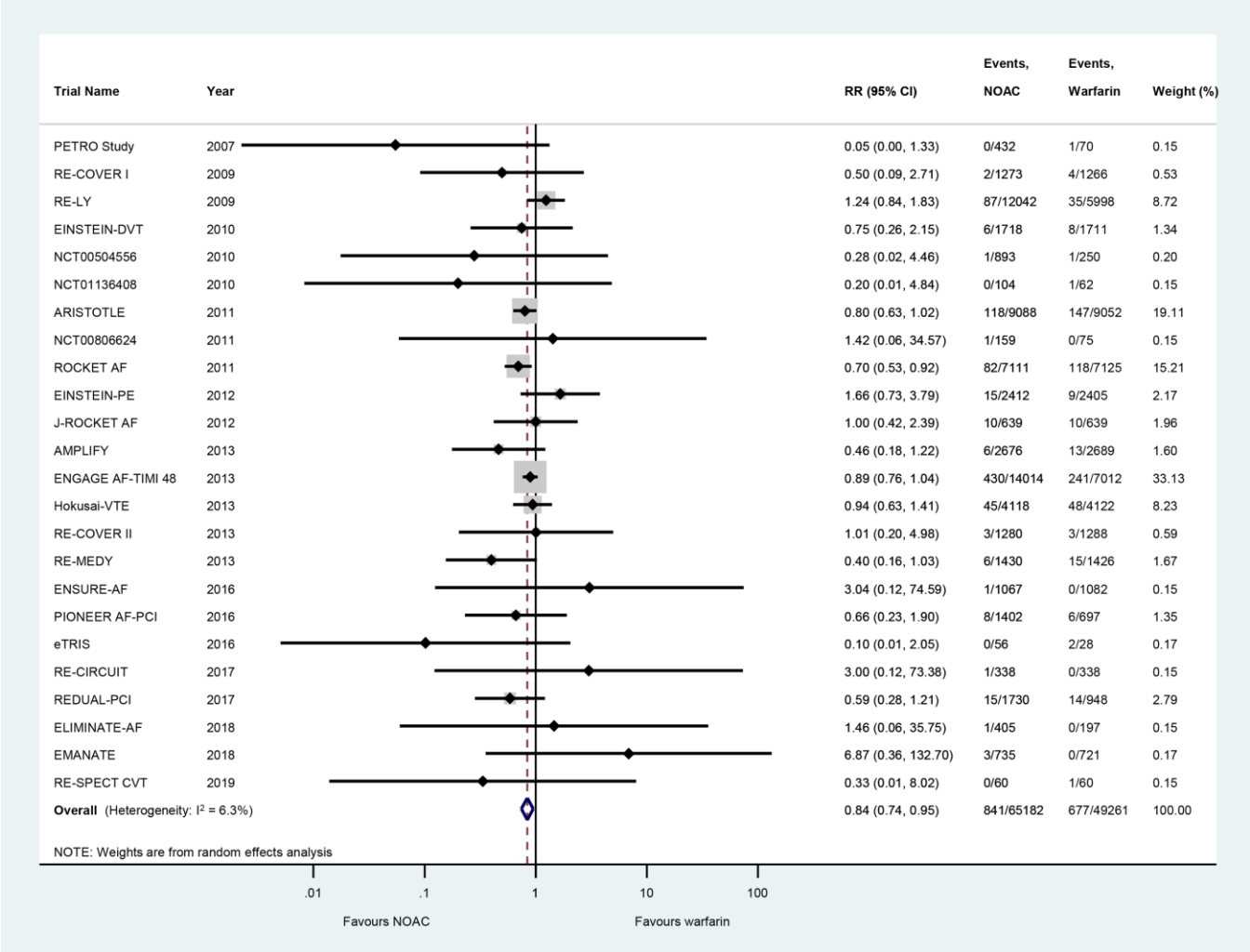
Abbreviations: CI, confidence interval

Figure S3. Forest plot of the relative risk of any fracture associated with NOACs compared to warfarin in the cohort studies subgroup



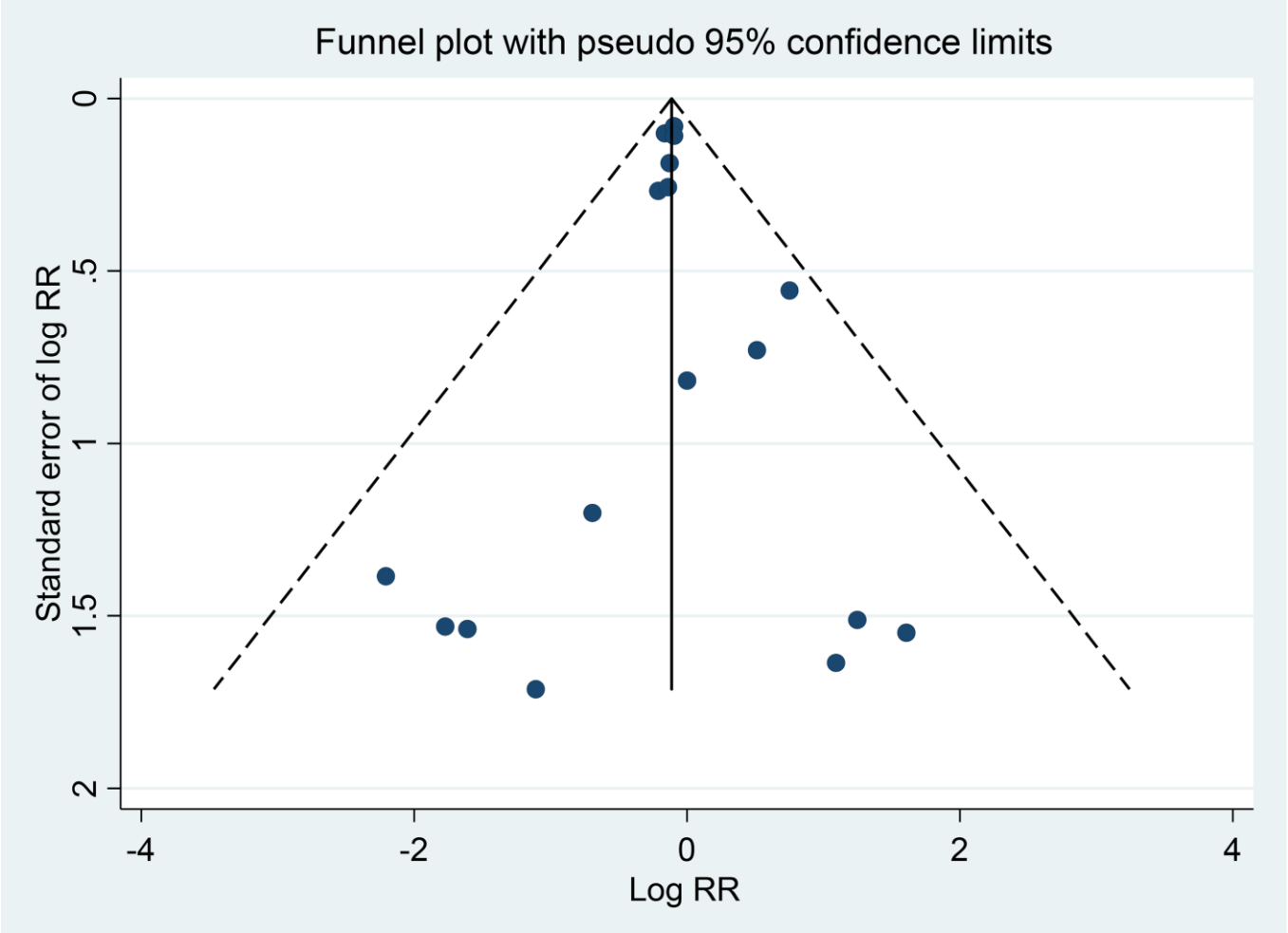
Abbreviations: CI, confidence interval; RR, relative risk

Figure S4. Forest plot of the relative risk of any fracture associated with NOACs compared to warfarin in the randomized controlled trials subgroup



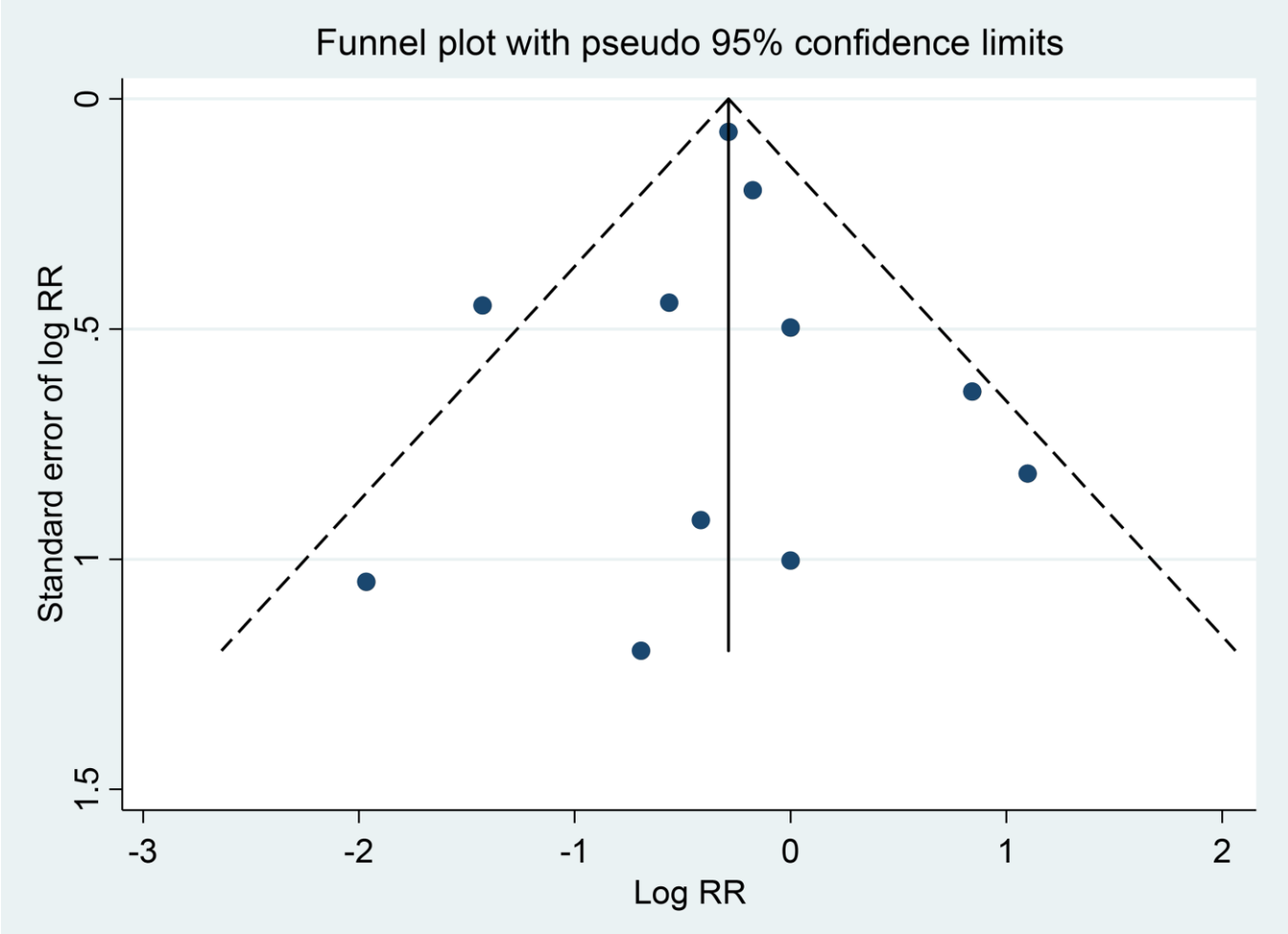
Abbreviations: CI, confidence interval; RR, relative risk

Figure S5. Funnel plot for the publication bias in studies comparing hip fracture risk between NOACs and warfarin



Abbreviations: RR, relative risk

Figure S6. Funnel plot for the publication bias in studies comparing vertebral fracture risk between NOACs and warfarin



Abbreviations: RR, relative risk